

## SAFETY DATA SHEET

Creation Date 10-Sep-2009

Revision Date 27-Mar-2024

Revision Number 4

### 1. Identification

**Product Name** Chlorobenzene

**Cat No. :** L09522

**CAS No** 108-90-7  
**Synonyms** Monochlorobenzene; Benzene chloride

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

Thermo Fisher Scientific Chemicals, Inc.  
30 Bond Street  
Ward Hill, MA 01835-8099  
Tel: 800-343-0660  
Fax: 800-322-4757

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### **Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

|                                    |            |
|------------------------------------|------------|
| Flammable liquids                  | Category 3 |
| Acute Inhalation Toxicity - Vapors | Category 4 |
| Skin Corrosion/Irritation          | Category 2 |

#### Label Elements

##### **Signal Word**

Warning

##### **Hazard Statements**

Flammable liquid and vapor  
Causes skin irritation  
Harmful if inhaled

**Precautionary Statements****Prevention**

Use only outdoors or in a well-ventilated area  
Wash face, hands and any exposed skin thoroughly after handling  
Wear protective gloves/protective clothing/eye protection/face protection  
Do not breathe dust/fume/gas/mist/vapors/spray  
Keep away from heat/sparks/open flames/hot surfaces. - No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Use explosion-proof electrical/ventilating/lighting equipment  
Use only non-sparking tools  
Take precautionary measures against static discharge

**Response**

Get medical attention/advice if you feel unwell

**Inhalation**

IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing  
Call a POISON CENTER or doctor/physician if you feel unwell

**Skin**

If skin irritation occurs: Get medical advice/attention  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/shower  
Wash contaminated clothing before reuse

**Eyes**

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
If eye irritation persists: Get medical advice/attention

**Fire**

In case of fire: Use CO<sub>2</sub>, dry chemical, or foam for extinction

**Storage**

Store in a well-ventilated place. Keep cool

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

Toxic to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component     | CAS No   | Weight % |
|---------------|----------|----------|
| Chlorobenzene | 108-90-7 | >95      |

### 4. First-aid measures

**General Advice**

If symptoms persist, call a physician.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if

symptoms occur.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**Most important symptoms and effects**

None reasonably foreseeable. Causes central nervous system depression: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Notes to Physician**

Treat symptomatically

## 5. Fire-fighting measures

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Unsuitable Extinguishing Media**

Water may be ineffective

**Flash Point**

23 °C / 73.4 °F

**Method -**

No information available

**Autoignition Temperature**

590 °C / 1094 °F

**Explosion Limits****Upper**

9.6 vol %

**Lower**

1.8 vol %

**Sensitivity to Mechanical Impact**

No information available

**Sensitivity to Static Discharge**

No information available

**Specific Hazards Arising from the Chemical**

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Phosgene. Hydrogen chloride gas.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
3

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions**

Use personal protective equipment as required. Ensure adequate ventilation.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Ensure adequate ventilation.

**Storage.**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Incompatible Materials. Strong oxidizing agents. Bases. Strong reducing agents. Metals.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component     | ACGIH TLV   | OSHA PEL                                                                                                   | NIOSH          | Mexico OEL (TWA)           |
|---------------|-------------|------------------------------------------------------------------------------------------------------------|----------------|----------------------------|
| Chlorobenzene | TWA: 10 ppm | (Vacated) TWA: 75 ppm<br>(Vacated) TWA: 350 mg/m <sup>3</sup><br>TWA: 75 ppm<br>TWA: 350 mg/m <sup>3</sup> | IDLH: 1000 ppm | TWA: 5 ppm<br>STEL: 15 ppm |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Use explosion-proof electrical/ventilating/lighting equipment. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

**Personal Protective Equipment****Eye/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Recommended Filter type:**

Organic gases and vapours filter. Type A. Brown. conforming to EN14387.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Liquid                   |
| Appearance                             | Clear                    |
| Odor                                   | bitter almonds           |
| Odor Threshold                         | No information available |
| pH                                     | No information available |
| Melting Point/Range                    | -45 °C / -49 °F          |
| Boiling Point/Range                    | 131 °C / 267.8 °F        |
| Flash Point                            | 23 °C / 73.4 °F          |
| Evaporation Rate                       | 1 (Butyl Acetate = 1.0)  |
| Flammability (solid,gas)               | Not applicable           |
| Flammability or explosive limits       |                          |
| Upper                                  | 9.6 vol %                |
| Lower                                  | 1.8 vol %                |
| Vapor Pressure                         | 12 mbar @ 20°C           |
| Vapor Density                          | 3.9                      |
| Specific Gravity                       | 1.108                    |
| Solubility                             | Moderately soluble       |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | 590 °C / 1094 °F         |
| Decomposition Temperature              | > 132°C                  |
| Viscosity                              | 0.8 mPa.s @ 20°C         |
| Molecular Formula                      | C6 H5 Cl                 |
| Molecular Weight                       | 112.56                   |

## 10. Stability and reactivity

**Reactive Hazard**

None known, based on information available

|                                         |                                                                                                       |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
| <b>Stability</b>                        | Stable under recommended storage conditions.                                                          |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Bases, Strong reducing agents, Metals                                        |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Phosgene, Hydrogen chloride gas              |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                              |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                         |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component     | LD50 Oral                      | LD50 Dermal                  | LC50 Inhalation              |
|---------------|--------------------------------|------------------------------|------------------------------|
| Chlorobenzene | LD50 2000 - 4000 mg/kg ( Rat ) | LD50 > 7940 mg/kg ( Rabbit ) | LC50 = 13.5 mg/L ( Rat ) 7 h |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                          |
|------------------------|--------------------------|
| <b>Irritation</b>      | Irritating to skin       |
| <b>Sensitization</b>   | No information available |
| <b>Carcinogenicity</b> |                          |

| Component     | CAS No   | IARC       | NTP        | ACGIH | OSHA       | Mexico |
|---------------|----------|------------|------------|-------|------------|--------|
| Chlorobenzene | 108-90-7 | Not listed | Not listed | A3    | Not listed | A3     |

ACGIH: (American Conference of Governmental Industrial Hygienists)

A1 - Known Human Carcinogen  
A2 - Suspected Human Carcinogen  
A3 - Animal Carcinogen  
ACGIH: (American Conference of Governmental Industrial Hygienists)  
Mexico - Occupational Exposure Limits - Carcinogens  
A1 - Confirmed Human Carcinogen  
A2 - Suspected Human Carcinogen  
A3 - Confirmed Animal Carcinogen  
A4 - Not Classifiable as a Human Carcinogen  
A5 - Not Suspected as a Human Carcinogen

Mexico - Occupational Exposure Limits - Carcinogens

|                                                   |                                                                                                                               |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Mutagenic Effects</b>                          | No information available                                                                                                      |
| <b>Reproductive Effects</b>                       | No information available.                                                                                                     |
| <b>Developmental Effects</b>                      | No information available.                                                                                                     |
| <b>Teratogenicity</b>                             | No information available.                                                                                                     |
| <b>STOT - single exposure</b>                     | None known                                                                                                                    |
| <b>STOT - repeated exposure</b>                   | None known                                                                                                                    |
| <b>Aspiration hazard</b>                          | No information available                                                                                                      |
| <b>Symptoms / effects, both acute and delayed</b> | Causes central nervous system depression: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Endocrine Disruptor Information</b>            | No information available                                                                                                      |

**Other Adverse Effects**

Tumorigenic effects have been reported in experimental animals.

## 12. Ecological information

**Ecotoxicity**

The product contains following substances which are hazardous for the environment. Contains a substance which is: Very toxic to aquatic organisms.

| Component     | Freshwater Algae                                                                                                                | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                                                                             | Microtox                                                                                                                          | Water Flea                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Chlorobenzene | EC50: = 12.5 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: 2.55 - 420 mg/L, 96h (Pseudokirchneriella subcapitata) | LC50: = 91 mg/L, 96h static (Brachydanio rerio)<br>LC50: 4.1 - 5.3 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 4.1 - 4.9 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 6.9 - 7.9 mg/L, 96h flow-through (Lepomis macrochirus)<br>LC50: 36.35 - 58.19 mg/L, 96h static (Poecilia reticulata)<br>LC50: = 4.5 mg/L, 96h static (Pimephales promelas)<br>LC50: 7 - 8.5 mg/L, 96h flow-through (Pimephales promelas) | EC50 = 11.26 mg/L 30 min<br>EC50 = 11.3 mg/L 30 min<br>EC50 = 11.5 mg/L 15 min<br>EC50 = 20 mg/L 10 min<br>EC50 = 9.36 mg/L 5 min | EC50: = 0.59 mg/L, 48h (Daphnia magna) |

**Persistence and Degradability**

Persistence is unlikely

**Bioaccumulation/ Accumulation**

No information available.

**Mobility**

. Will likely be mobile in the environment due to its water solubility.

| Component     | log Pow |
|---------------|---------|
| Chlorobenzene | 3.79    |

## 13. Disposal considerations

**Waste Disposal Methods**

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component                | RCRA - U Series Wastes | RCRA - P Series Wastes |
|--------------------------|------------------------|------------------------|
| Chlorobenzene - 108-90-7 | U037                   | -                      |

## 14. Transport information

**DOT**

UN-No UN1134  
Proper Shipping Name CHLOROBENZENE  
Hazard Class 3  
Packing Group III

**TDG**

UN-No UN1134  
Proper Shipping Name CHLOROBENZENE  
Hazard Class 3  
Packing Group III

**IATA**

UN-No UN1134  
Proper Shipping Name CHLOROBENZENE  
Hazard Class 3  
Packing Group III

**IMDG/IMO**

|                      |               |
|----------------------|---------------|
| UN-No                | UN1134        |
| Proper Shipping Name | CHLOROBENZENE |
| Hazard Class         | 3             |
| Packing Group        | III           |

**15. Regulatory information****United States of America Inventory**

| Component     | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|---------------|----------|------|-----------------------------------------------|-----------------------------|
| Chlorobenzene | 108-90-7 | X    | ACTIVE                                        | -                           |

**Legend:**

**TSCA** - US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component     | CAS No   | DSL | NDL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|---------------|----------|-----|-----|-----------|-------|------|------|------|-------|----------|
| Chlorobenzene | 108-90-7 | X   | -   | 203-628-5 | X     | X    | X    | X    | X     | KE-25489 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**U.S. Federal Regulations****SARA 313**

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372.

| Component     | CAS No   | Weight % | SARA 313 - Threshold Values % | SARA 313 - Reporting thresholds |
|---------------|----------|----------|-------------------------------|---------------------------------|
| Chlorobenzene | 108-90-7 | >95      | 1.0 %                         | -                               |

**SARA 311/312 Hazard Categories**

Should this product meet EPCRA 311/312 Tier reporting criteria at 40 CFR 370, refer to Section 2 of this SDS for appropriate classifications.

**CWA (Clean Water Act)**

| Component     | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|---------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Chlorobenzene | X                          | 100 lb                      | -                      | X                         |

**Clean Air Act**

| Component     | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|---------------|-----------|-------------------------|-------------------------|
| Chlorobenzene | X         |                         | -                       |

**OSHA** - Occupational Safety and Health Administration

Not applicable

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

| Component     | Hazardous Substances RQs | CERCLA Extremely Hazardous Substances RQs | SARA Reportable Quantity (RQ) |
|---------------|--------------------------|-------------------------------------------|-------------------------------|
| Chlorobenzene | 100 lb                   | -                                         | 100 lb<br>45.4 kg             |

**California Proposition 65** This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component     | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|---------------|---------------|------------|--------------|----------|--------------|
| Chlorobenzene | X             | X          | X            | X        | X            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

**U.S. Department of Homeland Security** This product does not contain any DHS chemicals.

**Other International Regulations**

**Mexico - Grade** Serious risk, Grade 3

**Authorisation/Restrictions according to EU REACH**

| Component     | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|---------------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Chlorobenzene | 108-90-7 | -                                                                   | Use restricted. See item 75.<br>(see link for restriction details)            | -                                                                                                     |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component     | CAS No   | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|---------------|----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Chlorobenzene | 108-90-7 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

**Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?**

Not applicable

**Other International Regulations**

| Component | CAS No | Seveso III Directive | Seveso III Directive | Rotterdam | Basel Convention |
|-----------|--------|----------------------|----------------------|-----------|------------------|
|-----------|--------|----------------------|----------------------|-----------|------------------|



|               |          | (2012/18/EC) -<br>Qualifying Quantities<br>for Major Accident<br>Notification | (2012/18/EC) -<br>Qualifying Quantities<br>for Safety Report<br>Requirements | Convention (PIC) | (Hazardous Waste) |
|---------------|----------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------|------------------|-------------------|
| Chlorobenzene | 108-90-7 | Not applicable                                                                | Not applicable                                                               | Not applicable   | Annex I - Y45     |

## 16. Other information

**Prepared By** Health, Safety and Environmental Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Creation Date** 10-Sep-2009  
**Revision Date** 27-Mar-2024  
**Print Date** 27-Mar-2024  
**Revision Summary** New emergency telephone response service provider.

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**